

Songeun Kim

ssekim.610@gmail.com [linkedin.com/in/ssekim610](https://www.linkedin.com/in/ssekim610) github.com/sok610 sskim610.vercel.app Irvine, CA

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL, C/C++, Ruby, Arduino C++

Frontend & Backend: React, TailwindCSS, HTMX, Django, Flask, Node.js, REST APIs, Celery, Redis, PostgreSQL, Ruby on Rails, Docker

AI / ML: Hugging Face Transformers, PyTorch, Embeddings, NLP, LLM workflows, OpenAI API, Gemini API

Data & Infra: ELK Stack, Kafka, InfluxDB, Grafana, AWS (EC2, S3, VPC), Nginx, MQTT

EXPERIENCE

Software Engineer

Jet Bridge LLC (automotive data & Digital Key platform, Hyundai/Kia)

Costa Mesa, CA

Oct 2025 – May 2026

- Developed a PoC LLM-powered pipeline using **OpenAI API** to extract and classify **8,000+** unstructured Jira support tickets — normalized noisy operational data into structured schemas and prototyped NLP embedding-based clustering with **Elasticsearch**.
- Designed an end-to-end daily snapshot system ingesting vehicle sales and Digital Key telemetry from TMS Oracle DB and **PostgreSQL**; built **React**-powered monitoring platform covering **10 pages** displaying DSL/**PostgreSQL** query results as tables and pie charts.
- Engineered operational web features using **React**, **Node.js**, and **Flask**: Digital Key history sorting, key status aggregation, and per-transaction API response time extraction; optimized **5M+** row queries from **~10 min to ~1 min**.
- Performed real-time incident monitoring during Kubernetes pod scaling events (10 → 12–14 pods); tracked exception types, API response times, and transaction throughput, reporting findings to senior engineers.

Software Engineer Intern

DT Master Carbon (ESG goal tracking platform)

Berkeley, CA

Jul 2025 – Sep 2025

- Reduced geospatial JSON data load time from **~1 min to ~3 sec (~95% improvement)** by replacing monolithic data fetches with on-demand API endpoints and server-side caching.
- Enhanced frontend usability by refactoring the UI with modern **JavaScript**, replacing legacy event handlers with a central state object — eliminating recurring UI errors and creating a more responsive, error-free user workflow.
- Engineered a dynamic filtering system in **Django** Admin using **HTMX**, reducing a selection list from **200+** items to under **10**; added Excel export validation (openpyxl) eliminating recurring data corruption incidents.

PROJECTS

Soundlog — AI-Powered Social Music Journaling Platform

Apr 2025 – Present

Django · PostgreSQL · Redis · Celery · Hugging Face · OpenAI API · Gemini API · Spotify API · AWS EC2/S3 · Nginx

github.com/sok610/soundlog github.com/sok610/soundlog-ai

- Engineered an automated batch script using **Gemini API** to generate one daily music-themed journaling prompt and persist it to **PostgreSQL**, driving user retention and structuring the core entry context for the AI recommendation system.
- Reduced p95 recommendation latency from **4.7s to 0.22s** via **Redis** caching (12h TTL), **Celery** pre-warming, and **PostgreSQL** connection pooling; fault-tolerant design with exponential backoff deployed on **AWS EC2/S3**.
- Integrated DistilRoBERTa emotion classification with embeddings and prompt-engineered recommendation workflows; designed architecture for a human-in-the-loop feedback optimization pipeline.

Cactus — Real-Time IoT Environmental Monitoring System

May 2026 – Present

Arduino C++ · ESP32 · MQTT · Kafka · InfluxDB · Grafana · Docker · AWS EC2

github.com/sok610/cactus-monitor

- Assembled and programmed ESP32 microcontrollers; wrote **Arduino C++** firmware streaming sensor data to **AWS EC2** via **MQTT**; containerized all components as **Docker** images; built pipeline: **Kafka** → **InfluxDB** → live **Grafana** dashboards.

PoliticianInfo — SaaS Platform for Political Transparency

Aug 2024 – Sep 2024

Ruby on Rails · Google Maps API · Google Civic API · Google News API · RSpec · Cucumber · Capybara

- Built a SaaS platform from scratch serving comprehensive information on **500+** politicians by state, with an interactive map powered by Google Maps API and real-time data enrichment via Google Civic and News APIs.
- Achieved **60%** test coverage with RSpec, Cucumber, and Capybara; designed relational schema and REST API layer supporting multi-source data aggregation.

EDUCATION

B.A. in Computer Science — Dean's Honor List, GPA: 3.74 / 4.00

Berkeley, CA

University of California, Berkeley

Aug 2021 – Aug 2024

Coursework: Data Structures, Algorithms, Software Engineering, AI, Machine Learning, Databases, Security, OOP, Computer Architecture